Importance of Maxwell's correction:

Earth receives radiations from Sun and other stars. These radiations travel through empty space where there are no electric charges and hence no electric current. Ampere's law says that only electric current can produce a magnetic field. If Ampere's law alone is true, there will not be any radiation.

Maxwell's correction term $\left(\mu_0 \epsilon_0 \frac{d\Phi_E}{dt}\right)$ in Ampere's law ensures that time–varying electric field or displacement current can also produce a magnetic field. Though conduction current is zero in an empty space, displacement current does exist. So, the equation (5.6) becomes

$$\oint_l \vec{B}.d\vec{l} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$$

In stars, due to thermal excitation of atoms, time–varying electric field is produced which in turn, produces time–varying magnetic field. According to Faraday's law, this time–varying magnetic field produces again time–varying electric field and so on. The coupled time–varying electric and magnetic fields travel through empty space with the speed of light and is called electromagnetic wave.

Even though Maxwell initially started with purely symmetry argument, his correction term explains one of the important aspects of the universe, namely the existence of electromagnetic waves.



Displacement current
The name stuck because Maxwell named it. The word displacement is poorly chosen because nothing is being displaced here.

EXAMPLE 5.1

Consider a parallel plate capacitor which is connected to an 230 V RMS value and 50 Hz frequency. If the separation distance between the plates of the capacitor and area of the plates are 1 mm and 20 cm^2 respectively. Calculate the displacement current at t = 1 s.

Solution

Potential difference between the plates of the capacitor,

$$V = V_{max} \sin 2\pi ft$$
$$= 230\sqrt{2}\ \sin(2\pi \times 50t)$$
$$\therefore V = 325 \sin 100\pi t$$

d = 1 mm = 1×10^{-3} m

A = 20 cm^2 = 20×10^{-4} m^2

Displacement current, $i_d = \epsilon_\circ \frac{d\Phi_E}{dt} = \epsilon_\circ \frac{d(EA)}{dt}$

$$\therefore i_d = \frac{\epsilon_0 A}{d}\left[\frac{dV}{dt}\right] \qquad \left[\because E = \frac{V}{d}\right]$$
$$= \frac{\epsilon_0 A}{d}(325)(100\pi)\cos 100\pi t$$
$$= \begin{pmatrix} 8.85\times10^{-12} \times 20 \times 10^{-4} \times 325 \\ \times 100 \times 3.14 \times \cos(100\pi \times 1) \end{pmatrix} \Big/ \left(1 \times 10^{-3}\right)$$
$$= 1.81 \times 10^{-6}\ \text{A} = 1.81\ \mu\text{A} \quad [\because \cos(100\pi \times 1) = 1]$$

5.1.2 Maxwell's equations in integral form

Electrodynamics can be summarized in four basic equations, known as Maxwell's equations. These equations are analogous to Newton's equations in mechanics. Maxwell's equations completely explain the behaviour of charges, currents and properties of electric and magnetic fields. These equations can be written in integral form (or integration form) or derivative form (or differential form). The

12th Physics_Vol-1_Unit 5_English Medium.indd 270 24-12-2021 15:57:27